Document 1: Project Planning & Management

Cosmetics Store Analytics & Business Intelligence System

Document: 01 - Planning & Management **Team Name:** Team 5

Team Members: Mahmoud Mohamed, Mahmoud Abdel Razek, Mahmoud Khairy, Adham, Youssef Tamer **Working Model:** Joint Collaborative Team

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1. Project Proposal

1.1 Project Overview

Retail stores handle high volumes of customer transactions every day, including product sales, quantity adjustments, and inventory restocking. In many small and medium retail businesses, sales records and stock lists are kept in separate files or recorded manually, making it difficult for store managers to track overall performance. This project builds a simple, reliable analytics and business intelligence system for a retail cosmetics store. The system collects raw transaction records, cleans and validates the data, stores it in a structured database, and produces clear summary reports to help managers track sales trends, popular categories, and stock levels.

1.2 Project Objectives

- Clean and standardize raw retail data by handling missing records, duplicate rows, and data type inconsistencies.
- Design and deploy a centralized relational database to store sales invoices, items, products, and customer accounts.
- Perform exploratory data analysis to calculate key retail metrics, including sales revenue, item volumes, and profit margins.
- Produce clear summary reports and business intelligence dashboards for retail decision-makers.

1.3 Project Scope

In-Scope:

- Processing historical sales invoices, line items, product catalogs, and account records.
- Database schema design, table creation, primary/foreign key relationships, and data loading in SQL Server.
- Data analysis using Python and SQL to evaluate sales patterns, customer segments, and inventory valuation.
- Creation of summary tables, trend charts, and academic project documentation.

Out-of-Scope:

- Direct integration with physical point-of-sale hardware, receipt printers, or barcode scanners.
- Building an online retail e-commerce website or digital payment processing gateway.
- Automated real-time purchase ordering with external cosmetic manufacturers and suppliers.

2. Project Plan

Note: Exact project start dates, durations, and deadlines have not been formally confirmed. The project is executed in a logical chronological sequence across the following stages.

2.1 Sequential Project Stages

Stage 1: Requirements & Planning — Define project scope, review retail requirements, and set team objectives.

Stage 2: Data Collection & Initial Inspection — Gather raw datasets and review data quality, structure, and missing fields.

Stage 3: Data Cleaning & Preprocessing — Clean raw records, validate relationships, remove duplicates, and prepare standardized files.

Stage 4: Database Design & ETL Implementation — Design relational schema, write table scripts, and load verified data into SQL Server.

Stage 5: Exploratory Data Analysis — Write Python analysis scripts to calculate sales metrics, profit margins, and inventory status.

Stage 6: Reporting & Dashboards — Build visual charts and summary tables presenting key store findings.

Stage 7: Testing & Verification — Verify mathematical totals, check database constraints, and ensure reproducibility.

Stage 8: Final Review & Presentation — Finalize documentation, prepare presentation materials, and complete project review.

2.2 Main Milestones

- **Milestone 1:** Project proposal, planning document, and requirements specification completed.
- **Milestone 2:** Cleaned datasets produced, audited, and reconciled against raw source records.
- **Milestone 3:** Relational database deployed with verified tables, foreign keys, and loaded data.
- **Milestone 4:** Exploratory data analysis completed with documented calculations and visual charts.
- **Milestone 5:** Final project documentation suite and presentation materials submitted.

2.3 Expected Deliverables

- Three core documentation files: Project Planning & Management, Literature Review, and Requirements Gathering.
- Python preprocessing scripts and verified cleaned CSV datasets.
- SQL database scripts including schema creation (DDL) and analytical queries (DML).
- Exploratory data analysis Jupyter Notebook containing structured analysis cells and charts.
- Project presentation slides summarizing project methodology, findings, and evaluation.

3. Task Assignment & Roles

Team 5 consists of five members: **Mahmoud Mohamed**, **Mahmoud Abdel Razek**, **Mahmoud Khairy**, **Adham**, and **Youssef Tamer**. All five members participate together as a joint collaborative team. Tasks across requirements gathering, data cleaning, database implementation, Python data analysis, testing, and documentation are shared collectively among all members without individual role division. The team holds regular study sessions to review code, verify calculations, and ensure that every member understands all aspects of the completed system.

4. Risk Assessment & Mitigation

The team identified potential project risks and developed practical solutions to avoid disruptions:

Risk Description	Potential Impact	Mitigation Strategy
Inconsistent or missing records in raw sales files	Calculation errors and distorted reporting results	Perform thorough data profiling and automated validation checks during preprocessing.
Delays in SQL database configuration or script errors	Delays in downstream data analysis and reporting	Design a simple normalized schema early and test loading scripts with sample tables first.
Software environment or Python library version differences	Code runs on one computer but fails on another	Use standard Python libraries and document all package requirements clearly.
Communication gaps or uneven workload distribution	Duplicate work or missed milestone deadlines	Hold regular team meetings, share files via version control, and review deliverables together.

5. Key Performance Indicators (KPIs)

To evaluate whether the project is executed successfully, the team defined the following project success measures (without inventing unverified numerical results):

1. **Data Reconciliation Accuracy:** 100% mathematical agreement between cleaned line-item sums and cleaned invoice totals.
2. **Database Relational Integrity:** Zero orphaned foreign keys and zero unhandled constraint errors in the SQL database.
3. **Sequential Milestone Completion:** Successful completion of all planned project stages in their proper sequence.
4. **Analysis Completeness:** Complete coverage of core retail analytics questions: revenue, item volume, profit margins, and stock status.
5. **Documentation Quality & Clarity:** Code and documentation are structured so that any university reviewer can reproduce the results.